Activity 6 Report: Database Backup, Restoration, and Recovery

COMP 018 Database Administration | Project tested: HabalGo (Ride and Booking Management System) (PHP + PostgreSQL) | Date of testing: 7 November 2026

1. Cover and links

Item	Value
Testing group	G08: Borreta, Joshua; Bola, Martin Lawrence; Juatchon, Joaquin Phillip; Marcos, Karl Raphael; Pacheco Jr., Sancho; Vera Cruz, Euan Joseph
Project tested	HabalGo (Ride and Booking Management System)
Owning group	Group 6 (member names are not listed in the repository)
Repository	https://github.com/jeanaroyo/HabalGo
Tracker	https://github.com/EclipseChan/bwisIT-Solutions---Campus-Canteen-Pre-Order-and-Queue-Management-System/tree/main/other_group_tests
Evidence files	a6_backup.txt, a6_restore.txt, a6_verify.txt, a6_access.txt, a6_recovery.txt

This is a re-test. The database was dropped and rebuilt from the repository files, and every result below comes from this run.

Run at about 00:24 on 7 November 2026 (Asia/Manila). The original database and the dump were not modified by the restore and recovery tests.

2. Backup plan

Item	Plan
Objects	Whole database grp00_habalgodb: 8 tables, 7 sequences, 17 indexes, 9 foreign keys, and the grants and default privileges
Method	pg_dump custom format (-Fc). The README does not describe any backup procedure, so this is the group's own plan
Account	The project defines no backup role, so the dump was taken as postgres. Recommend a role with pg_read_all_data
Files outside the database	The .env file (which the app does not read, see Activity 3) and whatever environment variables are used to start the server. The app stores no uploaded files
Schedule (real deployment)	Daily full dump at closing time, plus one before any schema change
Retention	7 daily and 4 weekly copies, one of them on another machine
RPO and RTO	RPO 24 hours (a day of bookings and payments could be lost), RTO 1 hour
Roles	Roles are not in a database dump. Keep pg_dumpall --roles-only next to each backup
Sensitive content	Customer names, phone numbers, emails, driver licence numbers and payments. The dump must be stored privately

3. Backup creation

```
pg_dump -U postgres -h localhost -Fc -d grp00_habalgodb -f grp00_habalgodb_activity6.dump
```

Item	Value
File	grp00_habalgodb_activity6.dump
Size	30,867 bytes
SHA-256	7B50FD5239BBEBC6CA42C3688732B1DB55C8E86F3D0BB4D316B37336D23AFCE0
Archive	custom format, gzip, 94 TOC entries, 8 TABLE DATA entries, 18 ACL entries and 2 DEFAULT ACL entries, dumped by pg_dump 18.6
Errors	None, exit code 0

4. Restoration and verification

```
createdb -U postgres -h localhost grp00_habalgodb_restore
pg_restore -U postgres -h localhost -d grp00_habalgodb_restore
grp00_habalgodb_activity6.dump
```

Restore succeeded with exit code 0 in 0.18 seconds and no messages.

Check	Original	Restored	Match
Tables, sequences, indexes	8, 7, 17	8, 7, 17	Yes
Row counts (booking, customer, driver, driver_vehicle, location, payment, rating_feedback, vehicle)	6, 11, 10, 0, 10, 2, 1, 1	same	Yes
Constraints: check, foreign key, not null, primary key, unique	4, 9, 33, 8, 4	same	Yes
Table grants per role (app_readonly_grp, app_runtime_grp, postgres)	8, 32, 56	same	Yes
md5 checksum of payment	a2788a29…13a2	a2788a29…13a2	Yes
md5 checksum of booking	db97ec5e…0e9e	db97ec5e…0e9e	Yes
Sequence positions (booking, customer, driver, location, payment, rating, vehicle)	10, 17, 11, 13, 8, 5, 4	same	Yes
Default privileges (pg_default_acl)	2 entries	2 entries, same ACLs	Yes
Check constraints still enforced (payment 0, rating 9)	rejected	rejected	Yes
Application started against the restored copy as habalgo_app_user	n/a	Dashboard showed 11 customers, 4 active bookings and revenue 250.5, and the bookings list returned 6, the same as the original	Yes

5. Access control after restore

Item	Original	Restored copy
Table privileges for app_runtime_grp and app_readonly_grp	32 and 8	32 and 8 (stored in the dump)
Default privileges for new tables and sequences	present	present, same ACLs
Database-level ACL (the CONNECT grants	app_runtime_grp=c, app_readonly_grp=c	empty (not restored)

from the script)		
habalgo_app_user reads customers, updates payment (no rows)	yes	yes, 11 customers
habalgo_readonly_user reads / inserts	reads / denied	reads / denied
Object owner	postgres	postgres

The roles themselves live in the cluster, so on this server they were still there and the grants lined up. On a new server they would have to be created first, or pg_restore prints errors for the grants. What did not come back is the database-level CONNECT grant. Because PUBLIC still has CONNECT on a normal database this causes no failure here, but it means the restored database does not carry the access plan the script set. After a restore on a new server, run 03_roles_and_privileges.sql again (it also creates the roles).

6. Recovery simulation

Step	Action	Result
Before	Checksum of payment in the restored copy	2 rows, md5 a2788a29…
Incident	UPDATE payment SET amount = 0.01, then DELETE of the Paid payments, directly in the database	UPDATE 2, DELETE 1. 1 row left, md5 96b6332a…, revenue as the dashboard reports it: 0
Recovery	createdb grp00_habalgodb_recovery and pg_restore of the same dump	Success, exit code 0
Compare	Checksum in the recovery database	2 rows, md5 a2788a29…, the same as before the incident; revenue back to 250.50
Original untouched	Checksum in grp00_habalgodb	2 rows, same md5

Outcome: recovered. A full restore into a new database was used because it leaves the damaged copy for comparison. Anything entered after the last backup would be lost, which is the RPO. Payments were chosen because the application account itself can edit and delete them (Activity 4), so this is the realistic way to lose money records.

7. Evaluation and recommendations

- The README has no backup or restore section. Add the dump and restore commands used here.
- Add pg_dumpall --roles-only to the procedure, and re-run the 03 script after a restore on a new server because the database-level grants do not come back.
- A daily dump is enough for a small project, but a day of bookings is what would be lost. If that is too much, use WAL archiving for point-in-time recovery.
- Store the dump on a different machine, keep the connection settings separately, and test a restore regularly.
- Taking away UPDATE and DELETE on payment from the application account (Activity 4) would make this kind of recovery less likely to be needed.

8. Limitations and errors

- A logical dump captures one moment only.
- The restore ran as postgres, so objects belong to postgres, the same as the original.
- The restore gave no errors or warnings.

- The test databases grp00_habalgodb_restore, grp00_habalgodb_recovery and grp00_habalgodb_alt are dropped after the evidence is captured.

9. Feedback for the owners

Backup and restore worked first time. Suggested additions: a backup section in the README, a roles-only dump, and re-running the roles script after restoring on another server.

AI usage declaration

- Tool: Claude (Anthropic).
- Tasks: interpreting the results of the testing.
- Validation: the commands were run by the group on its own computer, and every result in this report comes from the saved output files.